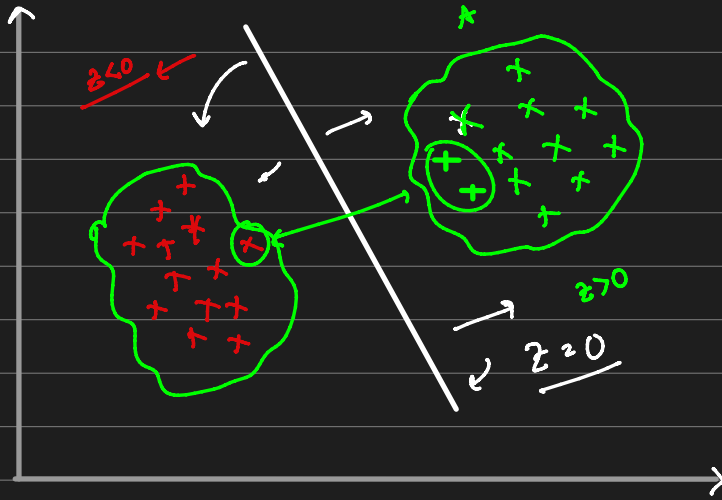


Support Vector Machine

Supervised
Classification
Regression.



100% consideration



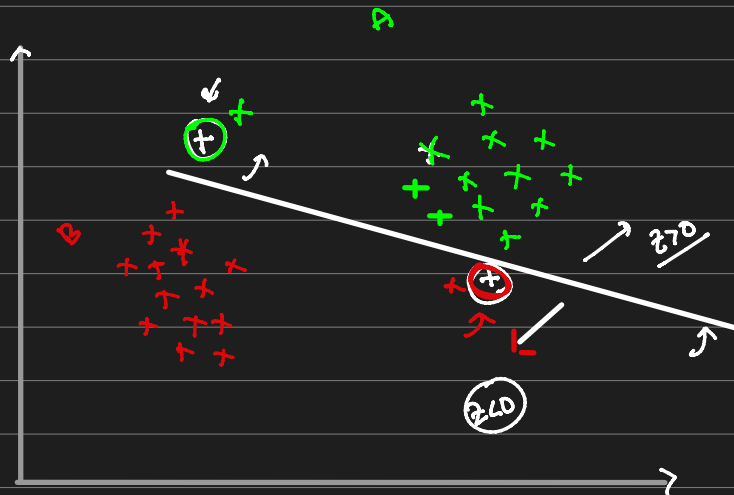
$z = wx + b$ → line, plane, hyperplane

$\sigma(z)$ = probability (1)

$\sigma(z) = \frac{1}{1 + e^{-z}}$

$z \geq 0$	$\sigma(0) = 0.5$	(Threshold)
$z < 0$	$\sigma(z) < 0.5$	(Negative)
$z > 0$	$\sigma(z) > 0.5$	(Positive)

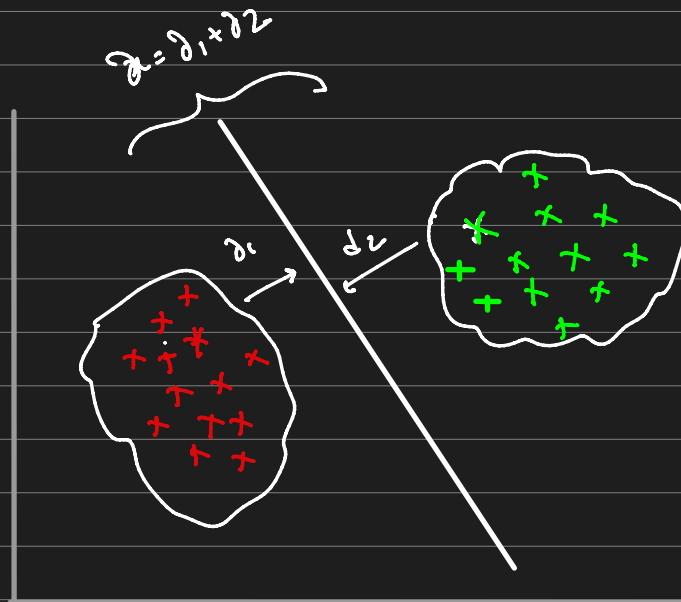
issue:



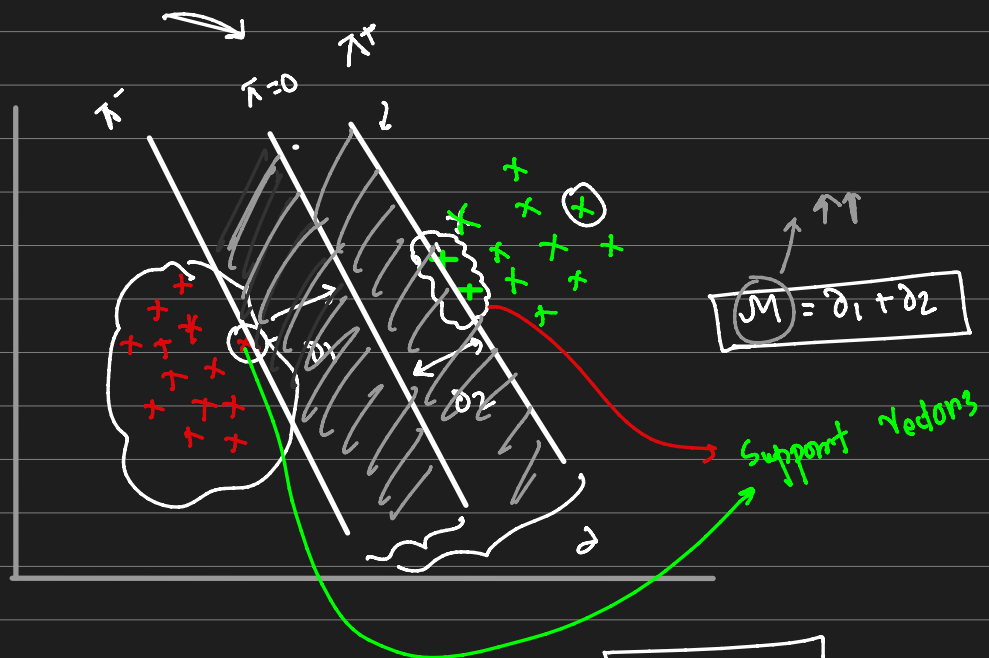
(SVM)

Margin = distance between classes

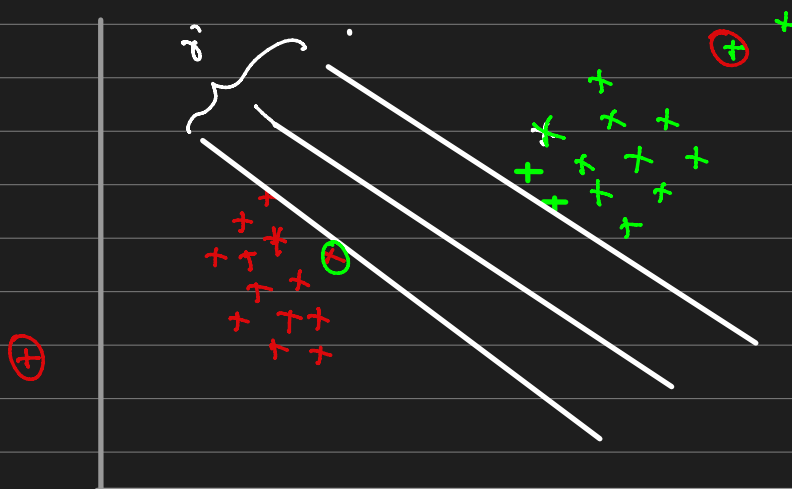
Maximum Margin Classifier



$$\pi = w \cdot x + b$$



$$d > d'$$



SVM

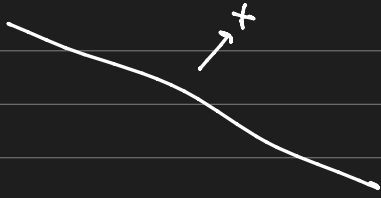
✓ (i) Robust to Outliers

✓ (ii) Generalized Model

✓ (iii) Kernel

$$\text{Margin} = d_1 + d_1 \\ = 2 \times d_1$$

$$d_1 = d_2$$



$$\text{distance} = \frac{w \cdot x + b}{\|w\|}$$

↑ score
↓ magnitude

$$\|w\|^2 = \left(\sqrt{w_1^2 + w_2^2 + \dots + w_n^2} \right)^2$$

$$d_1 = \frac{w \cdot x + b}{\|w\|} \rightarrow +1 \\ = \frac{1}{\|w\|}$$

$$\underline{\underline{M}} = d_1 + d_1$$

$$= 2 \times d_1$$

$$= 2 \times \frac{1}{\|w\|}$$

$$= \frac{2}{\|w\|}$$

$\|w\|$
 $2 \times \frac{1}{\|w\|}$
 $\frac{2}{\|w\|}$
 $\frac{2}{\|w\|}$

$$w = [w_1 \ w_2 \ w_3]$$

Target → Margin maximize

$$\frac{2}{2} = 1 \quad \frac{2}{4} = 0.5$$

$$=$$

$$\frac{2}{1} = 2 \quad \frac{2}{0.5} = 4 \quad \frac{2}{0.1} = 20$$

{ Hard Margin → zero points Margin এর মধ্যে
 আলাদা আলাদা

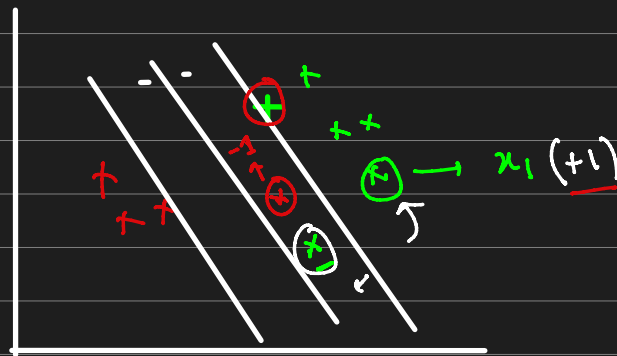
Margin $\uparrow \uparrow$

max = $\frac{2}{\|w\|}$ $\uparrow \uparrow$ $\rightarrow$ $\frac{\|w\|}{2}$ minimize

$\vec{w} = [w_1, w_2, w_3, \dots, w_n]$

$$J = \min \left(\frac{\|w\|^2}{2} + \sum \text{Hinge loss} \right)$$

Hinge loss = $\max \{0, 1 - y_i (wx + b)\}$



$x_1 \rightarrow (wx + b) \geq 1$

$H_L = \max(0, 1 - \boxed{y_i (wx + b)})$

$= \max(0, 1 - 2)$

$= \max(0, -1) = 0$

$\rightarrow (1 \times 2) = 2$

Correctly classified & distant from margin

$x_2 \rightarrow \boxed{wx + b} = 0.5$

$H_L = \max(0, 1 - \{+1 \times 0.5\})$

$= \max(0, 1 - 0.5) = \max(0, 0.5) = 0.5$

$\rightarrow$ Correctly classified but in the margin

$$x_2 \rightarrow wx+b = 0.5$$

$$H_2 = \max(0, 1 - \underline{y_i}(wx+b))$$

$$= \max(0, 1 - (-1) \times 0.5)$$

$$= \max(0, 1 - (-0.5))$$

$$= \max(0, 1+0.5) = \max(0, 1.5)$$

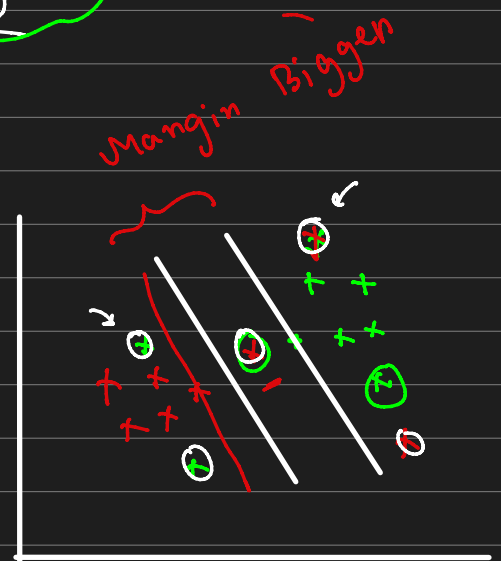
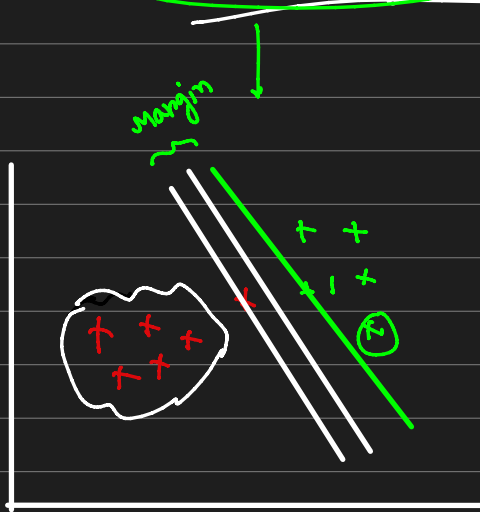
$$= \boxed{1.5}$$

misclassified
s in the
boundary

$$\left\{ \min \left\{ \underbrace{\frac{1}{2} \|w\|^2}_{\text{continuously}} + \sum_{i=1}^m \text{Hinge loss}(x_i) \right\} \right\}$$

→ Quadratic Pro (QP) optimization

Hard Margin & Soft Margin



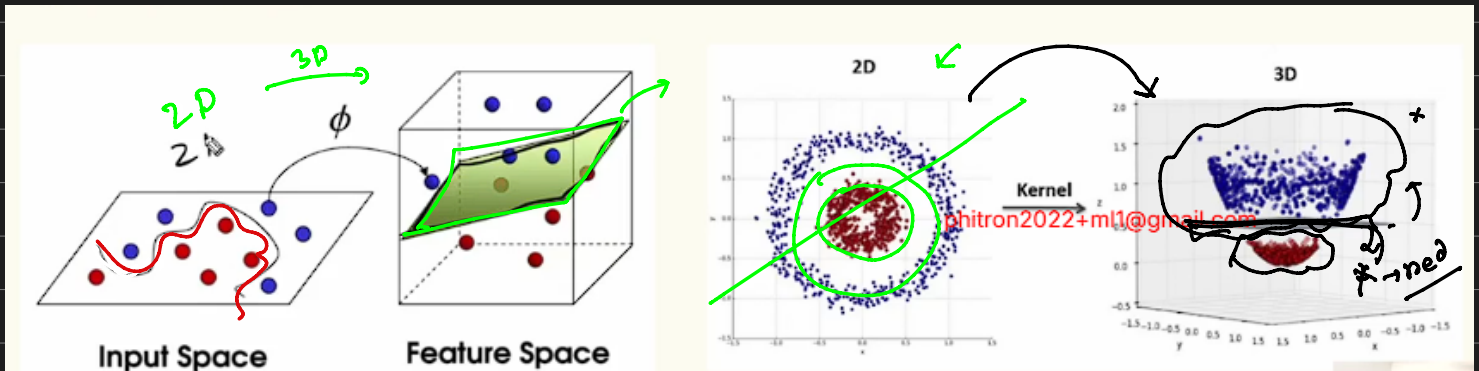
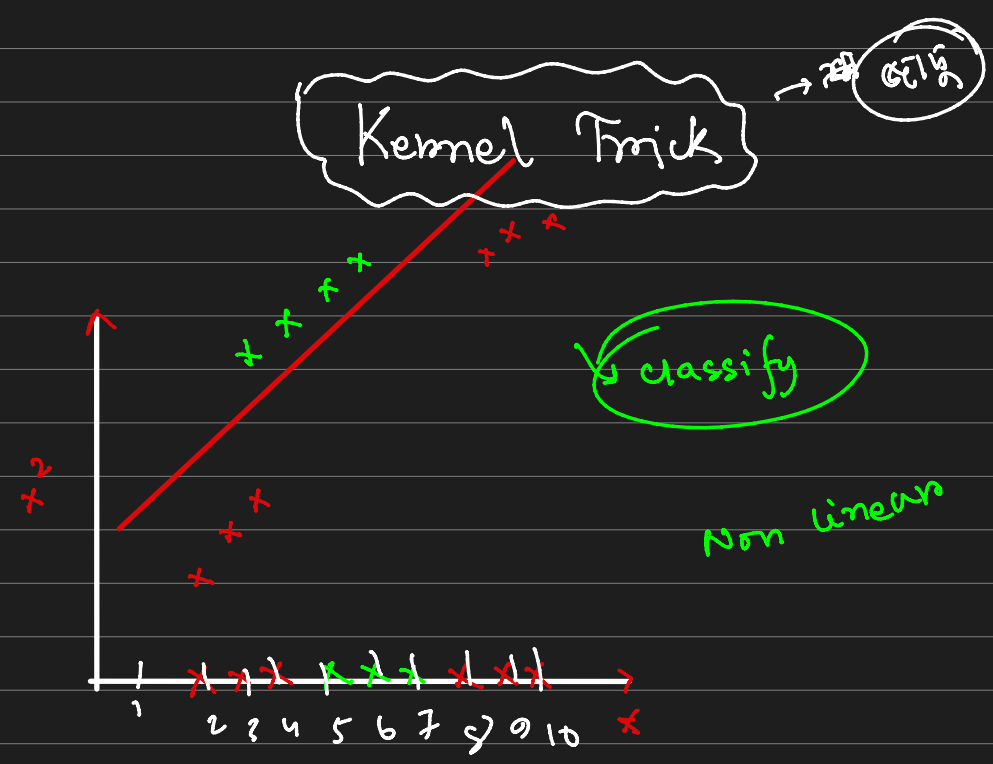
Hard Margin → Misclassification allowed or

Soft " → Misclassification allowed (some)

$$\min_{w, b} \left\{ \frac{\|w\|^2}{2} + C \sum_{i=1}^m \text{Hinge loss}(x_i) \right\}$$

The first term $\frac{\|w\|^2}{2}$ is labeled "Margin" and "margin squared".
 The second term $C \sum_{i=1}^m \text{Hinge loss}(x_i)$ is labeled "Penalty".
 The parameter C is labeled "Hyperparameter".

$C \uparrow$: Hinge LL $\rightarrow$ Misclassification $\downarrow$ $\rightarrow$ Hard Margin
 Margin $\downarrow$
 $C \downarrow$: Hinge LL $\rightarrow$ Margin $\uparrow$ $\rightarrow$ Soft margin



Kernel : ① linear } →
② Polynomial ✓
③ rbf kernel (Gaussian) ←

let's code Support Vector Classifier

Param:

$C \rightarrow C \uparrow \rightarrow$ Hard Margin
 $C \downarrow \rightarrow$ Soft Margin

kernel : 'linear', 'poly', 'rbf'

Gamma (for rbf)

degree (for poly)